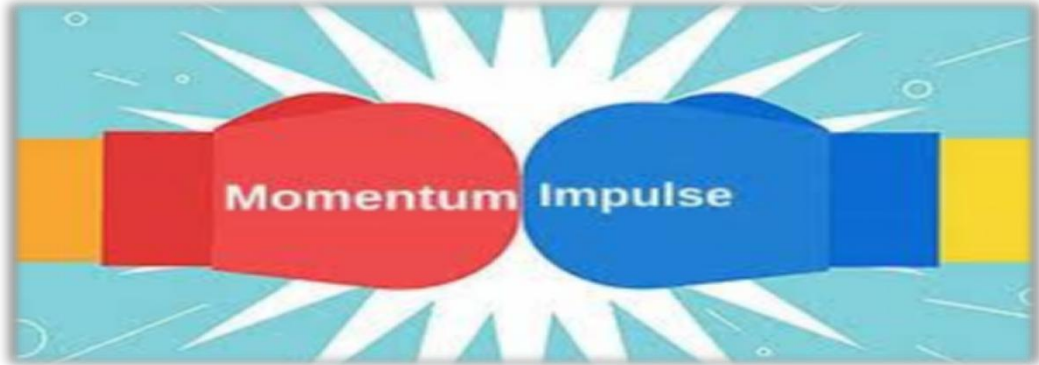
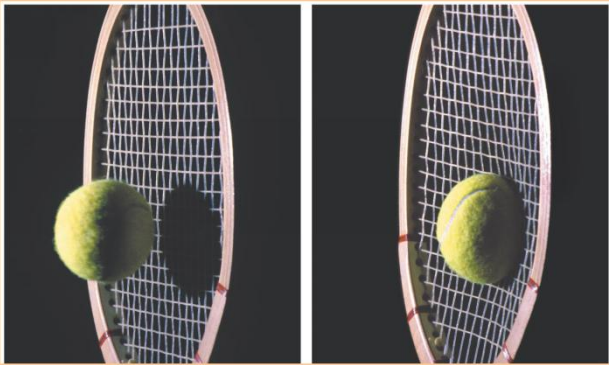




3.1 Momentum and Impulse

3.2 Conservation of Linear Momentum





3.1 MOMENTUM & IMPULSE

- (a) **Define** momentum & impulse $J = F\Delta t$
- (b) **Solve** problem related to impulse and impulse-momentum theorem

$$J = \Delta \vec{p} = m\vec{v}_f - m\vec{v}_i$$

- (c) **Interpret** impulse F - t graph



3.2 CONSERVATION OF LINEAR MOMENTUM

- (a) **State** the principle of conservation of linear momentum.
- (b) **Apply** the principle of conservation of momentum in elastic and inelastic collision in 1D
(Experiment 4 : Linear momentum)

OVERVIEW: Momentum & Impulse

Momentum & Impulse

Conservation of Linear
Momentum

Momentum

Impulse

Collision

Elastic

Inelastic

3.1 Momentum and impulse

Linear Momentum, p

Linear momentum, p of an object is defined as the **product of its mass and velocity**.

$$\vec{p} = m\vec{v}$$

SI unit of momentum : kg m s^{-1} or N s .

Momentum is **vector quantity** that has the **same direction as the velocity**.

Check Your Understanding

Which of the following statements are true about momentum?

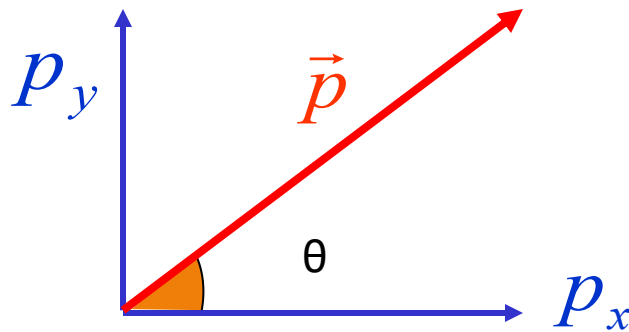
Momentum is a vector quantity.

The standard unit on momentum is the Joule.

An object with mass will have momentum.

An object with a changing speed will have a changing momentum.

It can be resolved into **vertical** (y-component) & **horizontal** (x-component).



$$p_x = p \cos \theta = mv \cos \theta$$

$$p_y = p \sin \theta = mv \sin \theta$$



Note:

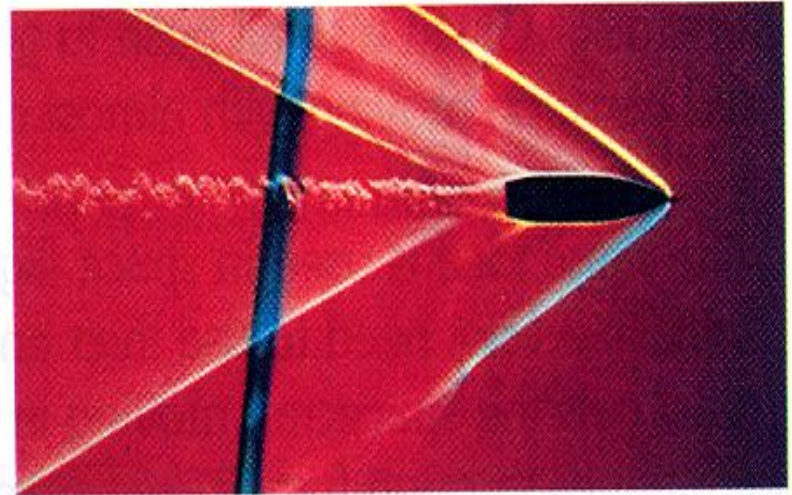
Momentum vector can be **resolved** into rectangular components and individual components can be then be added to find the components of the total momentum just as we do with force components.



The greater an object's momentum, the more force needed to stop it.



2000 kg moving at 5 m s^{-1}



0.02 kg moving at 400 m s^{-1}

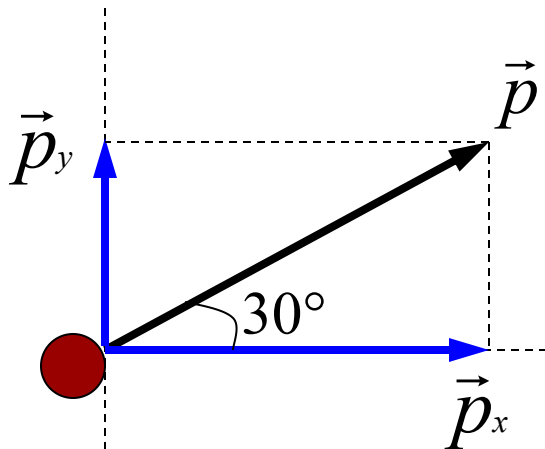
Both are hard to stop. The elephant has great mass, while single bullet has high velocity.

Example

A 1.5 kg ball was kicked with initial velocity of 40 m s^{-1} at the angle 30° with the horizontal line. Calculate the **momentum** of the ball and also the **horizontal and vertical components** of the momentum.

Solution

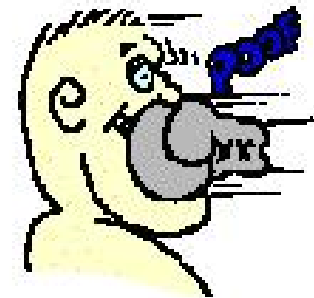
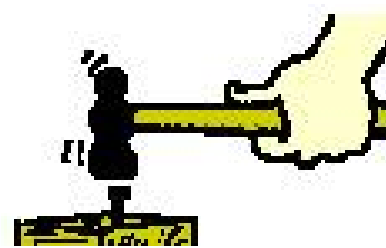
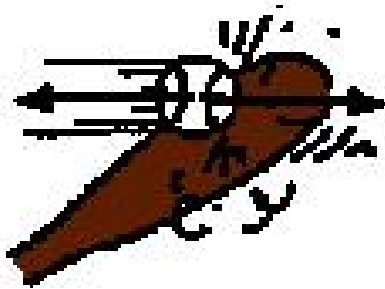
Momentum, $p = mv = 1.5(40) = 60 \text{ kg m s}^{-1}$
resolved into its components:



$$p_x = p \cos \theta = 60 \cos 30 = 51.96 \text{ kg ms}^{-1}$$

$$p_y = p \sin \theta = 60 \sin 30 = 30 \text{ kg ms}^{-1}$$

Impulse



When 2 objects – such as baseball and a bat, a hammer and a nail or even two cars – collide, they **exert large forces on one another for a short period of time, Δt .**

Forces of this type, which **exist only over a very short time during the collision** are called **impulsive force**.

Impulse

Defined as the **product of a force, F and the time, t**
OR the change of momentum.

It is a **vector quantity** whose **direction** is the **same**
as **the constant force** on the object

The S.I unit of impulse is **$N\ s$** or **$kgms^{-1}$**

Impulse – Momentum Theorem

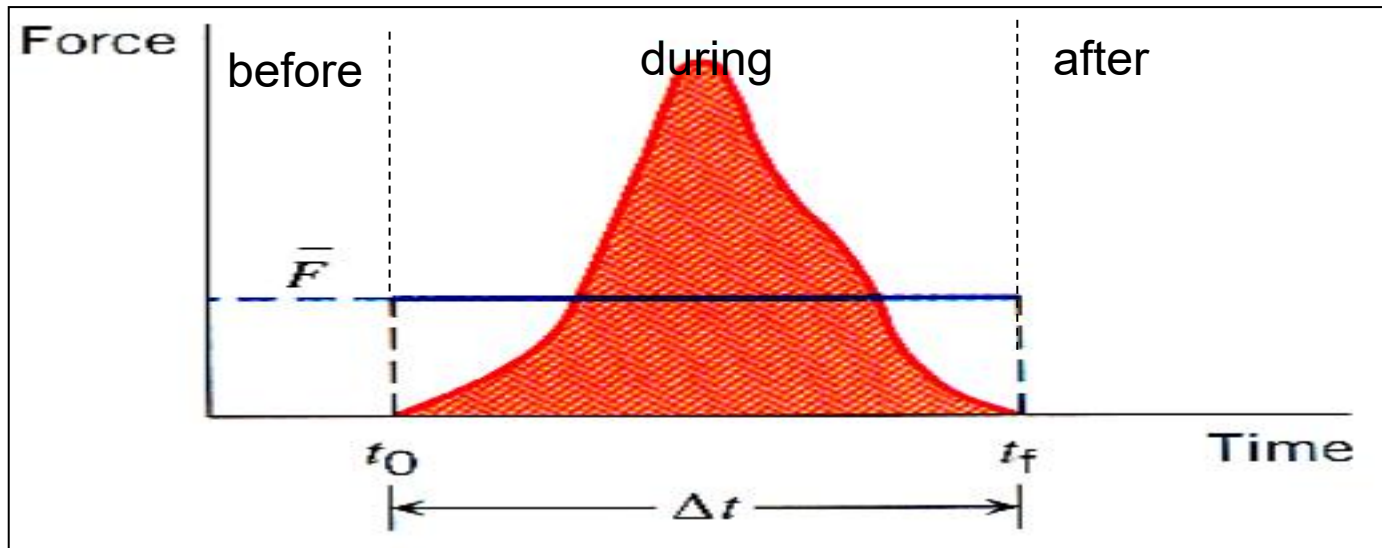
Impulse $\vec{J}$ of the force $\vec{F}$ equals to the *change in the momentum* of the particles.

$$\vec{J} = \Delta \vec{p}$$

where

$$\Delta \vec{p} = \vec{p}_{\text{final}} - \vec{p}_{\text{initial}}$$

$$\Delta \vec{p} = m\vec{v}_f - m\vec{v}_i$$



From Newton's 2nd law of motion

$$F = \frac{dp}{dt}$$

$$dp = Fdt$$

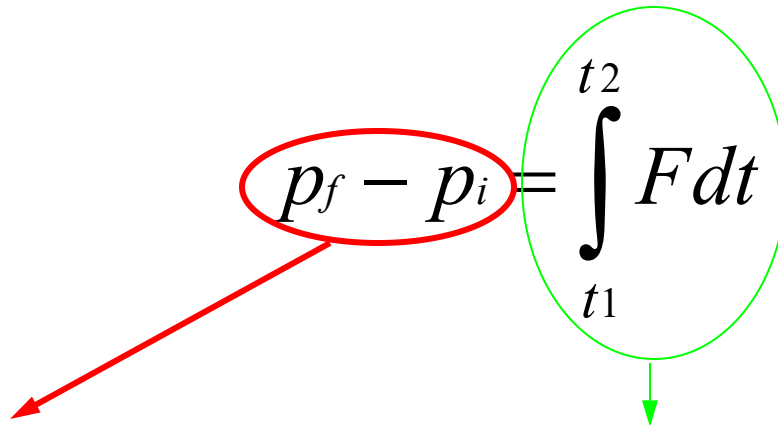
Integrating :

$$\int_{p_i}^{p_f} dp = \int_{t_1}^{t_2} F dt$$

where p_i – momentum of body at time t_i

p_f – momentum of body at time t_f



$$p_f - p_i = \int_{t_1}^{t_2} F dt$$


$\vec{J}$ = area under the F - t graph from t_i to t_f

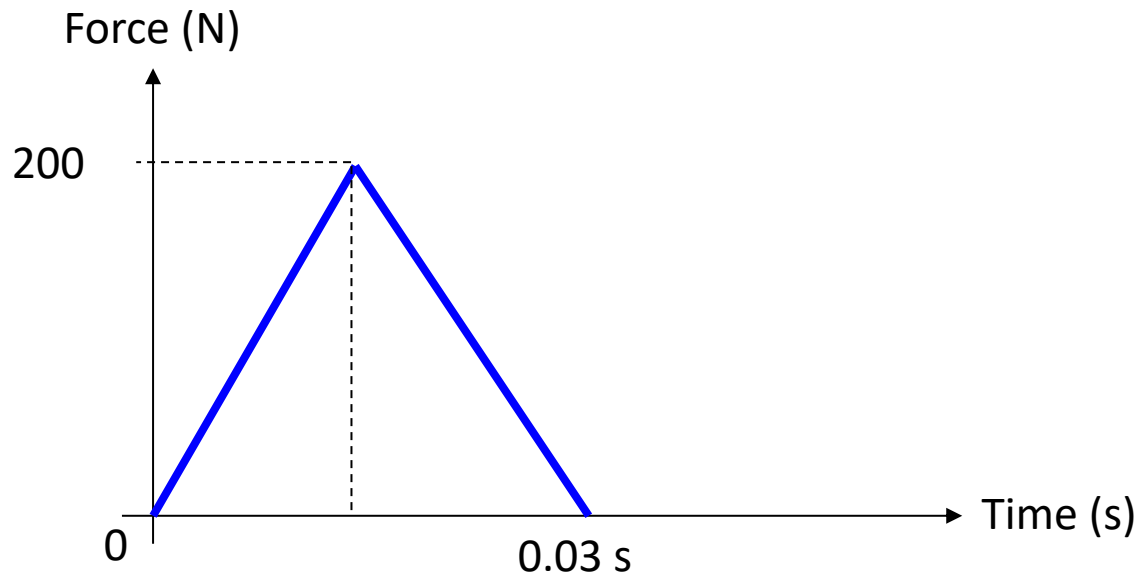
Impulse is equal to the **area under the F-t graph**.

if the force is **constant** (does not vary with time) or taking the average force for time interval Δt

$$\vec{J} = \bar{F} \Delta t$$

Example

The graph shows the force acting on a tennis ball of mass 60 g during a return shot.



- (a) What is the impulse on the ball?
- (b) If the ball reaches the player with velocity of 22 m s^{-1} moving to the left, what is the velocity of the return shot to the right?

Solution (Taking motion to the right as +ve)

(a) Impulse = area under the graph F-t

$$= \frac{1}{2} \times 0.03 \times 200$$

$$\vec{J} = 3 \text{ Ns}$$

$$(b) \vec{J} = \Delta \vec{p} = \vec{p}_{final} - \vec{p}_{initial}$$

$$= mv - mu$$

$$3 = 0.06(v - (-22))$$

$$v = 28 \text{ m s}^{-1} \quad (\text{to the right})$$

Example

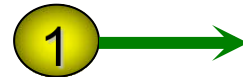
A 0.20 kg tennis ball strikes the wall horizontally with a speed of 100 m s^{-1} and it bounces off with a speed of 70 m s^{-1} in the opposite direction.

- Calculate the magnitude of impulse delivered to the ball by the wall,
- If the ball is in contact with the wall for 10 ms, determine the magnitude of average force exerted by the wall on the ball.

Solution :

$$m_1 = 0.20 \text{ kg}$$

$$u_1 = 100 \text{ m s}^{-1}$$



$$v_1 = 70 \text{ m s}^{-1}$$



Wall (2)

$$v_2 = u_2 = 0$$

Solution :

a. From the equation of impulse that the force is constant,

$$J = dp = p_2 - p_1$$

$$J = m_1(v_1 - u_1)$$

$$J = (0.20)[(-70) - 100]$$

$$J = -34 \text{ N s}$$

Therefore the magnitude of the impulse is **34 N s**.

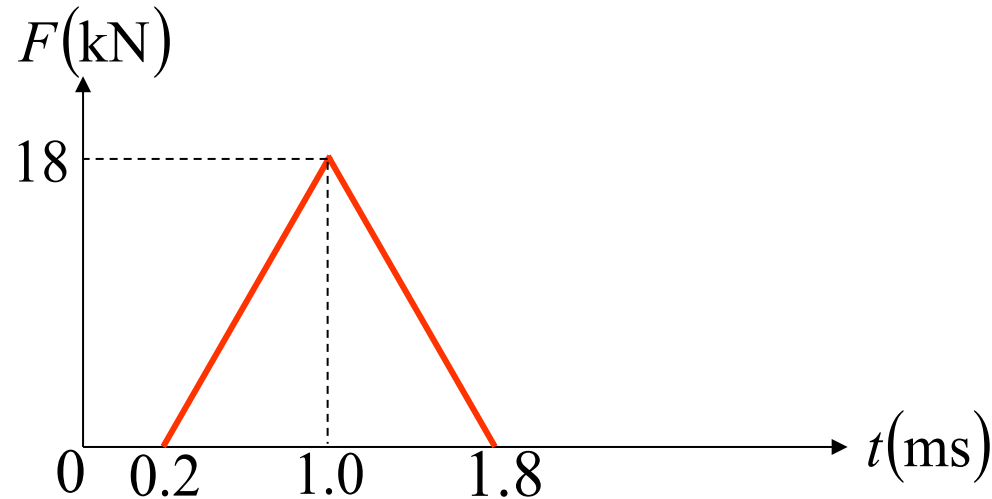
b. Given the contact time, $dt = 10 \times 10^{-3} \text{ s}$

$$J = F_{av} dt$$

$$34 = F_{av} (10 \times 10^{-3})$$

$$F_{av} = \mathbf{3400 \text{ N}}$$

Example



An estimated force-time curve for a tennis ball of mass 60.0 g struck by a racket is shown in Figure 3.21. Determine

- the impulse delivered to the ball,
- the speed of the ball after being struck, assuming the ball is being served so it is nearly at rest initially.

Solution

$$m = 60.0 \times 10^{-3} \text{ kg}$$

a. From the force-time graph,

J = area under the $F - t$ graph

$$J = \frac{1}{2}(1.8 - 0.2) \times 10^{-3} (18 \times 10^3)$$

$$J = 14.4 \text{ N s}$$

b. Given the ball's initial speed, $u = 0$

$$J = dp = m(v - u)$$

$$14.4 = (60.0 \times 10^{-3})(v - 0)$$

$$v = 240 \text{ m s}^{-1}$$

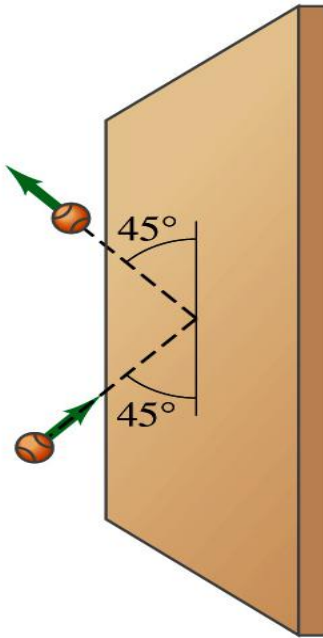
Follow up exercise

1. A steel ball with mass 40.0 g is dropped from a height of 2.00 m onto a horizontal steel slab. The ball rebounds to a height of 1.60 m.
 - a. Calculate the impulse delivered to the ball during impact.
 - b. If the ball is in contact with the slab for 2.00 ms, determine the average force on the ball during impact.

ANS. : 0.47 N s; 237.1 N

2. A golf ball ($m = 46.0$ g) is struck with a force that makes an angle of 45° with the horizontal. The ball lands 200 m away on a flat fairway. If the golf club and ball are in contact for 7.00 ms, calculate the average force of impact. (neglect the air resistance.)

ANS. : 293 N



3. A tennis ball of mass, $m = 0.060 \text{ kg}$ and a speed, $v = 28 \text{ m s}^{-1}$ strikes a wall at a 45° angle and rebounds with the same speed at 45° as shown. Calculate the impulse given by the wall.

ANS. : 2.4 N s to the left or -2.4 N s

Solution :

a) Given : $m = 40 \times 10^{-1} \text{ kg}$

height, $s_1 = -2\text{m}$

height, $s_2 = 1.6\text{m}$

$u = 0 \text{ ms}^{-1}$

$$\begin{aligned} v_1^2 &= u_1^2 - 2gs_1 \\ &= -2(9.81)(-2) \\ &= 6.26 \text{ms}^{-1} \end{aligned}$$

$$\begin{aligned} v_2^2 &= u_2^2 - 2gs_2 \\ &= -2g(1.6) \\ &= 5.6 \text{ms}^{-1} \end{aligned}$$

$$\begin{aligned}v_t &= v_1 + v_2 \\&= 6.26 + 5.6 \\&= 11.86 \text{ms}^{-1}\end{aligned}$$

$$\begin{aligned}I &= m(v - u)^0 \\&= 40 \times 10^{-3}(11.86) \\&= 0.474 \text{kgms}^{-1}\end{aligned}$$

b)

$$\begin{aligned}I &= (F_{\Delta v})(\Delta t) \\0.474 &= F_{\Delta v}(2 \times 10^{-3}) \\F_{\Delta v} &= 237 \text{N}\end{aligned}$$

2. using

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$200 = \frac{u^2 \sin 90^\circ}{9.81}$$

$$u = 44.29 \text{ m s}^{-1}$$

from

$$J = F_{\Delta v} \Delta t \dots \dots (1)$$

from

$$J = m\vec{v}^0 - mu$$

$$J = -mu \dots \dots (2)$$

$$u \rightarrow (2)$$

$$J = -mu$$

$$J = -(46 \times 10^{-3})(44.29)$$

$$= -2.04 \text{Ns}$$

$$J = \Delta t F_{\Delta v}$$

$$-2.04 = F_{\Delta v}(7 \times 10^{-3})$$

$$F_{\Delta v} = -291 \text{N}(\text{downward})$$

3. Given : mass, $m = 0.06\text{kg}$

$$u = 28\text{ms}^{-1}$$

$$v = 28\text{ms}^{-1}$$

$$\theta = 45^\circ$$

	x-axis	y-axis
u	$28 \sin 45^\circ$ $= 19.799$	$28 \cos 45^\circ$ $= 19.799$
v	$-28 \sin 45^\circ$ $= -19.799$	$28 \cos 45^\circ$ $= 19.799$

$$\begin{aligned}J_x &= m(v_x - u_x) \\&= 0.06[(-19.799) - (19.799)] \\&= -2.376Ns\end{aligned}$$

$$\begin{aligned}J_y &= m(v_y - u_y) \\&= 0.06(19.799 - 19.799) \\&= 0Ns\end{aligned}$$

$$\begin{aligned}J &= \sqrt{J_x^2 + J_y^2} \\&= \sqrt{(-2.375)^2 + (0)^2} \\&= 2.375Ns\end{aligned}$$

3.2 Conservation of Linear Momentum

Principle of Conservation of Linear Momentum

When the **net external force** acting on a system is **zero** (**isolated system**), the **total momentum** of that system is **constant** (or conserved).

$$\Sigma F_{net} = 0 \Rightarrow \frac{dp}{dt} = 0 \Rightarrow \vec{p}_{system} = \text{constant}$$

Or For an isolated system:

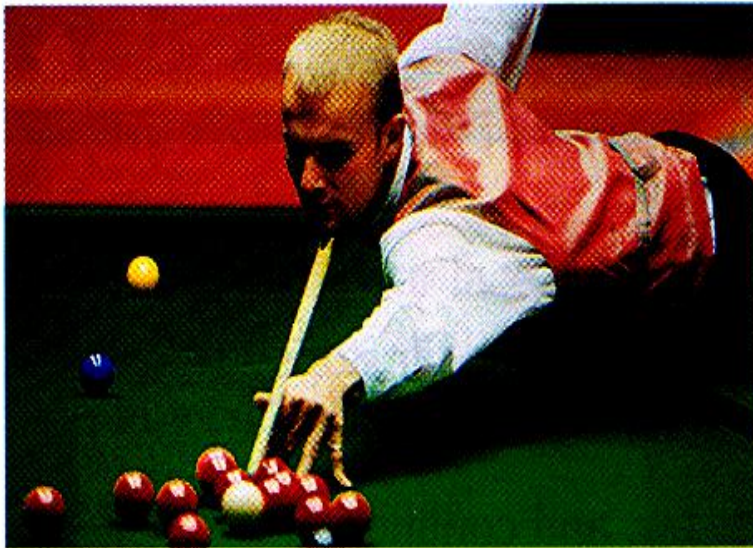
$$\Sigma \vec{p}_{initial} = \Sigma \vec{p}_{final}$$

"Conserved"
means
**"constant" or "not
changing."**





We often use the principle of conservation of momentum to solve problems where objects collide or explode apart.



Snooker players use the conservation of momentum instinctively with every shot



The downward momentum of the hot gases equals the upward momentum of the rocket

Collision

is an isolated event in which 2 or more bodies (the colliding bodies) exert relatively strong forces on each other for a relatively short time.

In a collision, the **change in the momentum of the system is zero** because no external forces are acting on the system.

Momentum is conserved for any collision

$$\sum \vec{p}_{\text{initial}} = \sum \vec{p}_{\text{final}}$$

There are 2 types of collision :

- (1) **Elastic**
- (2) **Inelastic**



In an elastic Collision,

Momentum and kinetic energy are conserved.

$$\sum \vec{p}_{\text{initial}} = \sum \vec{p}_{\text{final}}$$

$$\Sigma K_{\text{initial}} = \Sigma K_{\text{final}}$$

In an inelastic Collision,

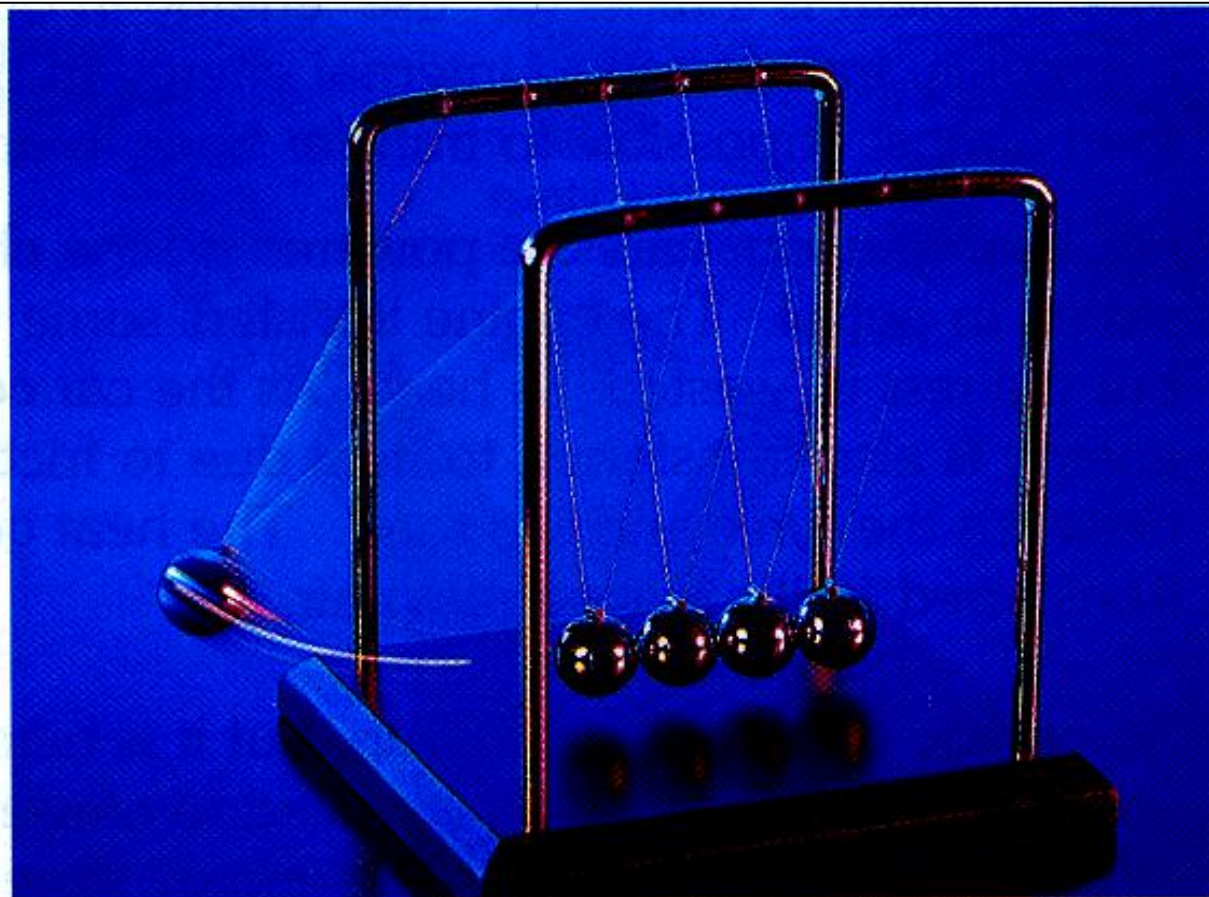
The momentum of a system is conserved.

$$\sum \vec{p}_{\text{initial}} = \sum \vec{p}_{\text{final}}$$

But its kinetic energy is not conserved.

$$\Sigma K_{\text{initial}} \neq \Sigma K_{\text{final}}$$

(part of the initial kinetic energy would have been transformed to other forms of energy like sound, heat & deformation.)



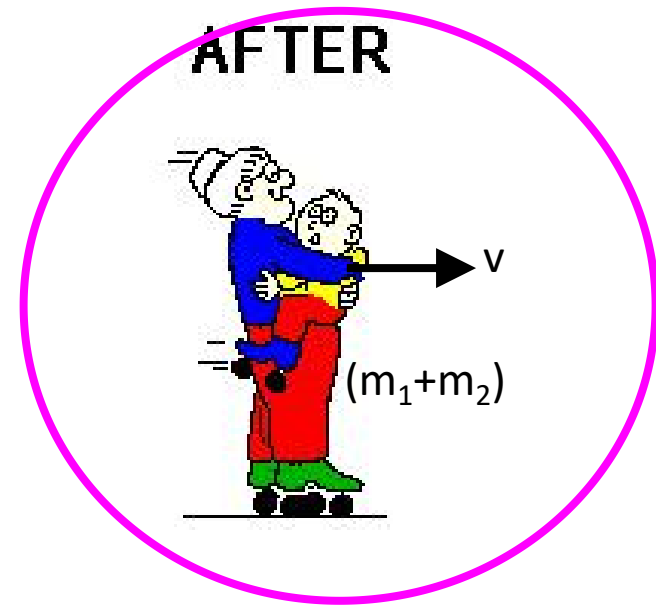
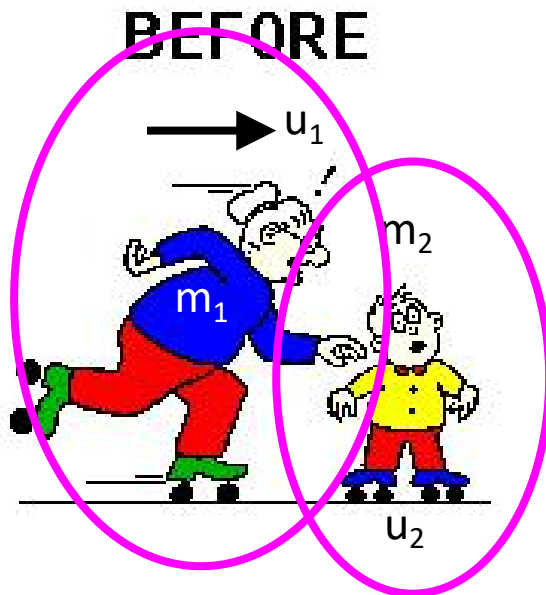
In this Newton's cradle the moving ball collides elastically with a stationary one of the same mass. The total KE and momentum are transferred from ball to ball.



In an inelastic collision, KE is 'lost'

Perfectly inelastic collision

For completely / perfectly inelastic collision, objects **stick together after impact**. The objects have the **same final velocity**.



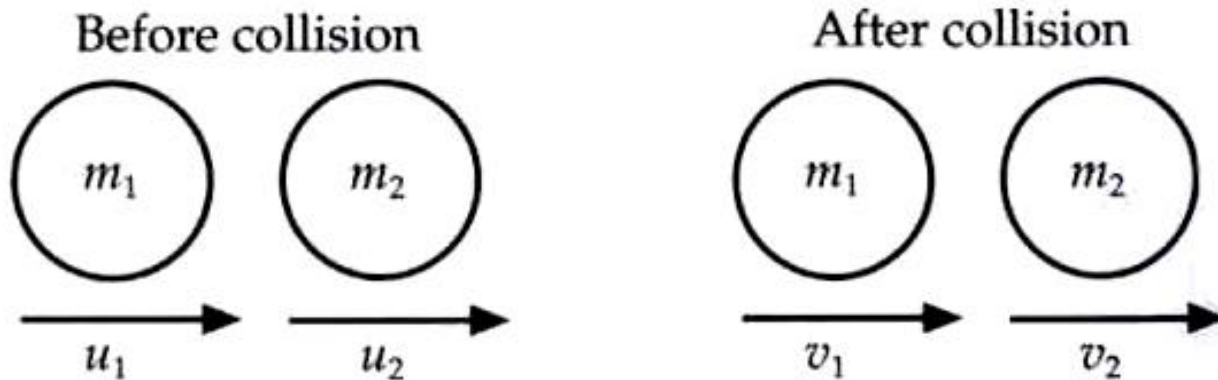
Apply Principle Conservation of Momentum,

$$\sum \vec{p}_{\text{before}} = \sum \vec{p}_{\text{after}}$$

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

Elastic Collision in ONE dimension motion

Consider 2 particles that undergo an **elastic head-on collision**.



In this case, both momentum and kinetic energy are conserved. According to the Principle of Conservation of Momentum:

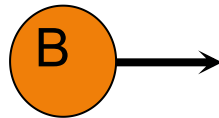
$$\sum \vec{p}_{\text{before}} = \sum \vec{p}_{\text{after}}$$
$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Knowing that kinetic energy is **conserved** for the system:

$$\Sigma K_{\text{before}} = \Sigma K_{\text{after}}$$
$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

Example**Linear momentum in one dimension collision**

$$u_B = 3 \text{ m s}^{-1}$$



$$u_A = 6 \text{ m s}^{-1}$$

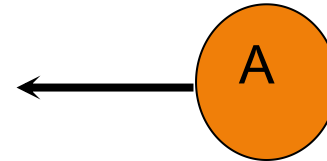


Figure above shows an object A of mass 200 g collides head-on with object B of mass 100 g. After the collision, B moves at a speed of 2 m s^{-1} to the left. Determine the velocity of A after collision.

Solution

$$m_A = 0.200 \text{ kg}; m_B = 0.100 \text{ kg}; u_A = -6 \text{ m s}^{-1}$$

$$u_B = 3 \text{ m s}^{-1}; v_B = -2 \text{ m s}^{-1}$$

$$\sum \vec{p}_i = \sum \vec{p}_f$$

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$(0.200)(-6) + (0.100)(3) = (0.200)v_A + (0.100)(-2)$$

$$v_A = -3.5 \text{ m s}^{-1} \text{ to the left}$$

Example

Object A of mass 8 kg moving at 4 m s^{-1} collides with another object B of mass 6 kg moving at 5 m s^{-1} in the opposite direction. After the collision, object A moves opposite to its initial direction at 0.1 m s^{-1} .

- (i) What is the velocity of B after the collision?
- (ii) Show that the collision is inelastic.

Solution

Before collision :

$$m_A = 8 \text{ kg} , u_A = 4 \text{ m s}^{-1}$$

$$m_B = 6 \text{ kg} , u_B = -5 \text{ m s}^{-1} \text{ (opposite direction with object A)}$$

after collision :

$v_A = -0.1 \text{ m s}^{-1}$ (opposite to its initial direction) ; $v_B = ?$

(i) Using Principle of Conservation of Momentum:

$$\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$$

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$8(4) + 6(-5) = 8(-0.1) + 6v_B$$

$$6v_B = 2.8$$

$$v_B = 0.467 \text{ m s}^{-1}$$

(ii) For an inelastic collision, kinetic energy of the system is not conserved

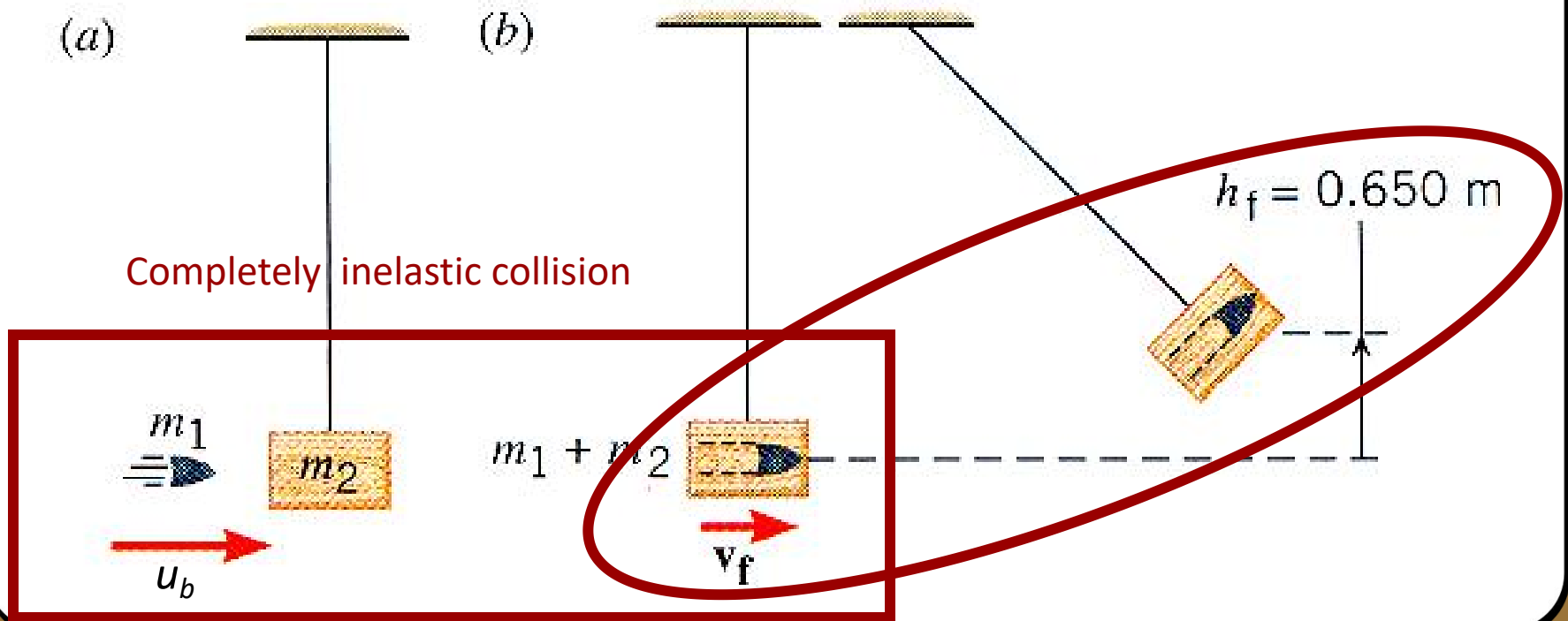
$$\begin{aligned} KE_{\text{before}} &= \frac{1}{2} m_A u_A^2 + \frac{1}{2} m_B u_B^2 \\ &= \frac{1}{2} (8)(4)^2 + \frac{1}{2} (6)(-5)^2 \\ &= 139 J \end{aligned}$$

$$\begin{aligned} KE_{\text{after}} &= \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 \\ &= \frac{1}{2} (8)(-0.1)^2 + \frac{1}{2} (6)(0.467)^2 \\ &= 0.6943 J \end{aligned}$$

$KE_{\text{before}} \neq KE_{\text{after}} \rightarrow \text{inelastic collision}$

Example

A stationary 2.50 kg block of wood suspended by a wire of negligible mass. A 0.01 kg bullet is fired into the block. The block (with the bullet in it) swings to a maximum height of 0.65 m above the initial position. Find the initial speed the bullet is fired.



Solution

Given: $m_1 = 0.01 \text{ kg}$, $m_2 = 2.50 \text{ kg}$

$h_f = 0.65 \text{ m}$

before collision, block is stationary $\rightarrow u_2 = 0$

find $u_1 = ?$

Completely inelastic collision

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v_f$$

$$0.01(u_1) + 0 = (0.01 + 2.50)v_f$$

$$u_1 = 251v_f \quad \cdots (1)$$

Block & bullet swings to a maximum height of 0.65 m. KE change to PE.

Apply Conservation of Mechanical energy

$$KE = PE$$

$$\frac{1}{2}(m_1 + m_2)v_f^2 = (m_1 + m_2)gh_f$$

$$\frac{1}{2}(2.51)v_f^2 = 2.51(9.81)(0.65)$$

$$v_f = 3.57 \text{ m s}^{-1}$$

Substitute v_f into (1) :

$$u_1 = 251(3.57) = 896.07 \text{ m s}^{-1}$$

Follow up exercise

1

On a touchdown attempt, a 95 kg running back runs towards the end zone at 3.75 m s^{-1} . A 111 kg linebacker moving at 4.10 m s^{-1} meets the runner in a head on collision. If the two players stick together,

- (a) What is their velocity immediately after collision?
- (b) What are the initial and final kinetic energies of the system?

Answer :

(a) -0.48 m s^{-1} ;

(b) $K_i = 1600 \text{ J}$; $K_f = 23.7 \text{ J}$

Solution

1. Given : $m_1 = 95 \text{ kg}$
 $m_2 = 111 \text{ kg}$
 $u_1 = 3.75 \text{ ms}^{-1}$
 $u_2 = -4.10 \text{ ms}^{-1}$

a)

$$\Sigma p_i = \Sigma p_f$$

$$m_1 u_i + m_2 u_2 = (m_1 + m_2) v$$

$$95(3.75) + 111(-4.10) = (95 + 111)v$$

$$v = -0.48 \text{ ms}^{-1}$$

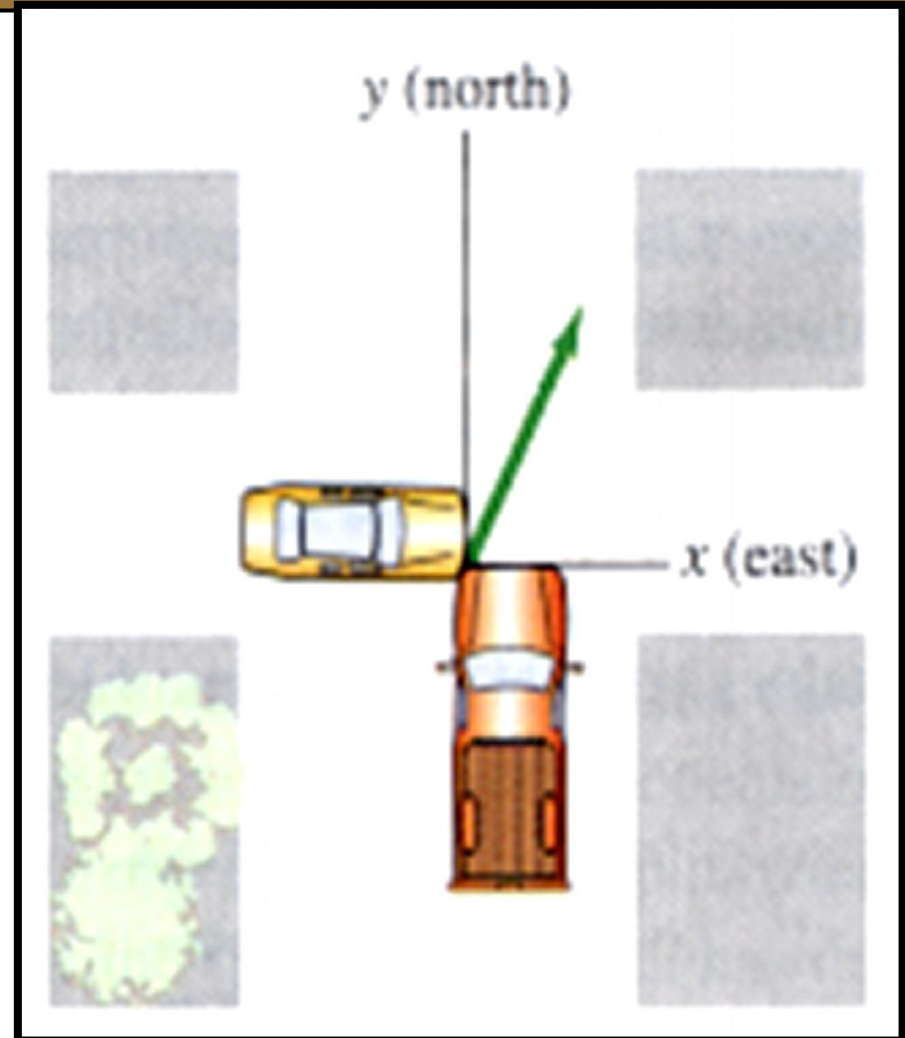
b)

$$\begin{aligned} KE_{initial} &= \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 \\ &= \frac{1}{2}(95)(3.75)^2 + \frac{1}{2}(111)(-4.10)^2 \\ &= 1600.93J \end{aligned}$$

$$\begin{aligned} KE_{final} &= \frac{1}{2}m_1v^2 + \frac{1}{2}m_2v^2 \\ &= \frac{1}{2}(95)(-0.48)^2 + \frac{1}{2}(111)(-0.48)^2 \\ &= 23.73J \end{aligned}$$

2

A car with mass of 950 kg and speed of 16 m s^{-1} approaches an intersection, as shown in figure below. A 1300 kg minivan traveling at 21 m s^{-1} is heading for the same intersection. The car and minivan collide and stick together. Find the speed and direction of the wrecked vehicles just after the collision.



Answer : 14 m s^{-1} at 61° above $+x$

2. Given : $m_1 = 950 \text{ kg}$
 $m_2 = 1300 \text{ kg}$
 $u_2 = 21 \text{ ms}^{-1}$

Horizontal motion, (x)

$$\Sigma p_i = \Sigma p_f$$

$$m_1 u_1 = (m_1 + m_2) v_x$$

$$950(16) = (950 + 1300) v_x$$

$$v_x = 6.76 \text{ ms}^{-1}$$

Vertical motion, (y)

$$\Sigma p_i = \Sigma p_f$$

$$m_2 u_2 = (m_1 + m_2) v_y$$

$$(1300)(21) = (950 + 1300) v_y$$

$$v_y = 12.13 \text{ ms}^{-1}$$

$$v = \sqrt{(v_x)^2 + (v_y)^2}$$

$$= \sqrt{(6.76)^2 + (12.13)^2}$$

$$= 13.88 \text{ ms}^{-1}$$

Direction,

$$\begin{aligned}\tan \theta &= \left(\frac{v_y}{v_x} \right) \\ &= \left(\frac{12.13}{6.78} \right)\end{aligned}$$

$$\theta = 61^0 \quad \text{Above } +x - \text{axis}$$

Follow up exercise

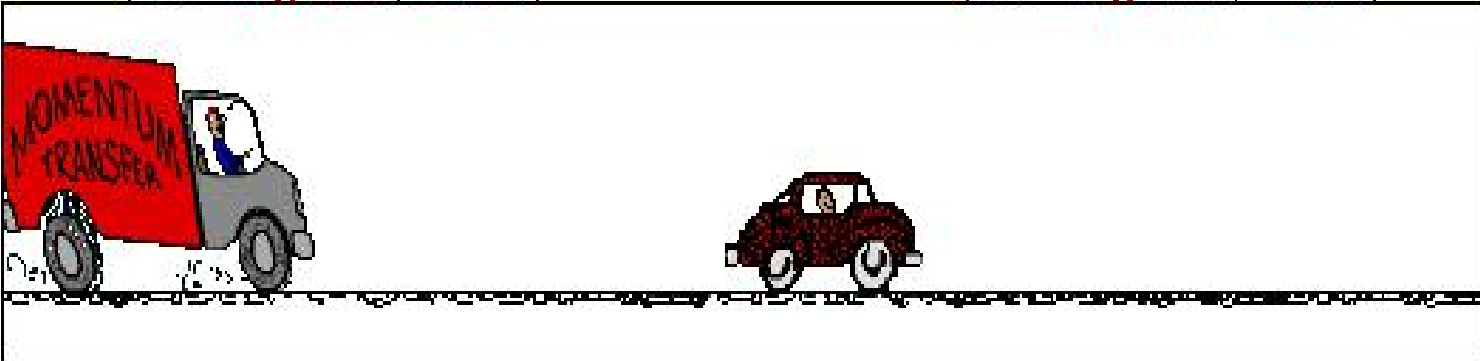
A karate student tries to break a board.



How can the abrupt stop of the hand generate so much force to break the board?

Assume that the hand has a mass of 0.35 kg and that the speeds of the hand just before and just after hitting the board are 10 m s^{-1} and 0 m s^{-1} respectively. What is the average force exerted by the fist on the board if the contact time is only 0.30 ms ?

Truck		Car	
mass (kg)	3000	mass (kg)	1000
vel. (m/s)	20.0	vel. (m/s)	0.0
mom. (kg m/s)	60 000	mom. (kg m/s)	0



End of Chapter 3